

GOA Pacific cod assessment

ESR

Ecosystem Status Report: Gulf of Alaska 2025



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With contributions from:



ESP

Ecosystem & Socioeconomic Profile (ESP): Gulf of Alaska (GOA) Pacific Cod

Kalei Shotwell & Russel Dame,
February 2026 SSC

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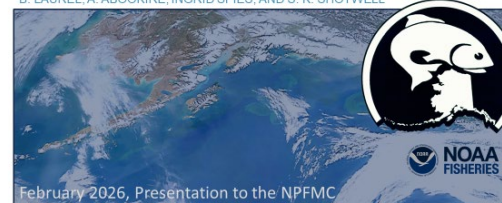


SAFE



GOA PACIFIC COD

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February 2026, Presentation to the NPFMC

Ecosystem Status Report: Gulf of Alaska 2025

Bridget Ferriss

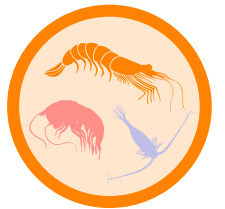
Alaska Fisheries Science Center, NOAA Fisheries

[full ESR attached to e-agenda: list of contributors p.1; Ecosystem Assessment p.4]



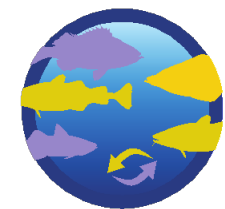
1. Warm at surface and depth, winter, spring and fall

- *[winter spawning; egg/larval survival; incr. demand for prey]*



2. Similarities to 2014-2016 in reduced lower trophic level productivity

- *[less productive food web to directly and indirectly support cod prey]*



3. Mixed upper trophic & ecosystem metrics

- *[cod body condition below average but some prey base and upper trophic indicators ok]*



4. Cooling in 2026 but concerns of residual heat & cumulative ecosystem effects

- *[elevated concerns of cumulative ecosystem response to warm year]*

Ecosystem and Socioeconomic Profile REPORT CARD SUMMARY



PREDICTIVE

Spawning Habitat Suitability: DECREASED TO LOW

- 🌡️ Warmest temperature at depth since 2019, negative effect on egg survival, potentially poor 2025 year class

CONTEXTUAL

🐟 RECRUITMENT

- Spring larvae & summer YOY CPUE **below average**; potentially poor 2025 year class

🌐 ECOSYSTEM

- Overall average status: **mixed** directional trends from 2024 to 2025 conditions

🌊 ENVIRONMENTAL

- Increase to **average** heatwave events (below 2015/16/19 levels); **decrease** to below **average** eddy kinetic energy (reduced larval retention/transport).

🐟 PREY

- Average spring bloom, sufficient forage for seabirds; decrease to below average Pacific cod **body condition**, slightly increased metabolic demand

🦈 PREDATOR

- Below average biomass consumed of Pacific cod (arrowtooth, cod, pollock multispecies model); steadily increasing Steller sea lion population in GOA

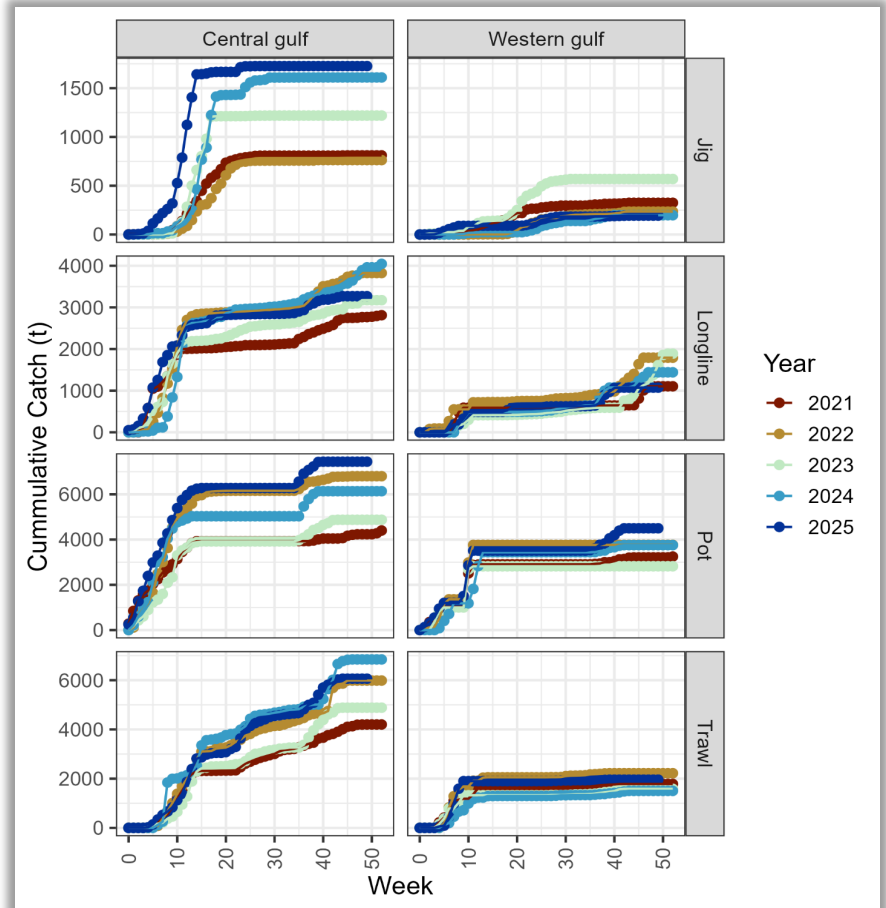
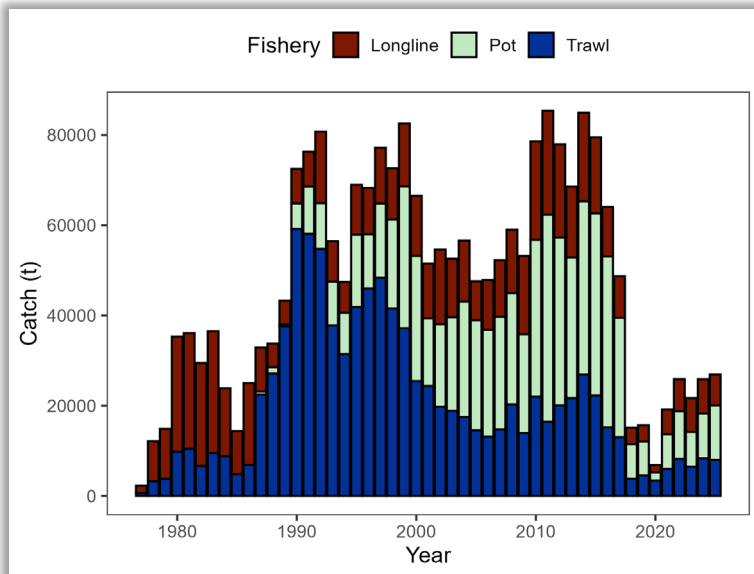
GOA COD ASSESSMENT OVERVIEW

- **BLUF:**
 - GOA Pacific cod: Tier 3b
 - 2025 estimated spawning biomass to be at $B_{34.3\%}$, projected in 2026 to be at $B_{33.1\%}$
 - 2026 recommended ABC (41,520 t) is a 29% increase from 2025 ABC (32,141 t)
 - Do not recommend reduction from maximum ABC
- From PT/SSC presentations, highlight:
 - Important data and risk table considerations

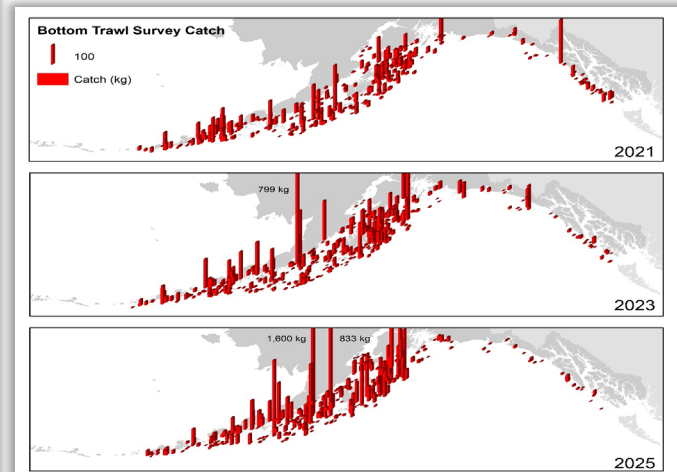
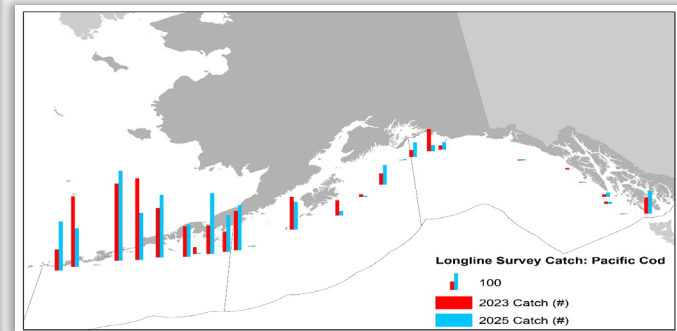
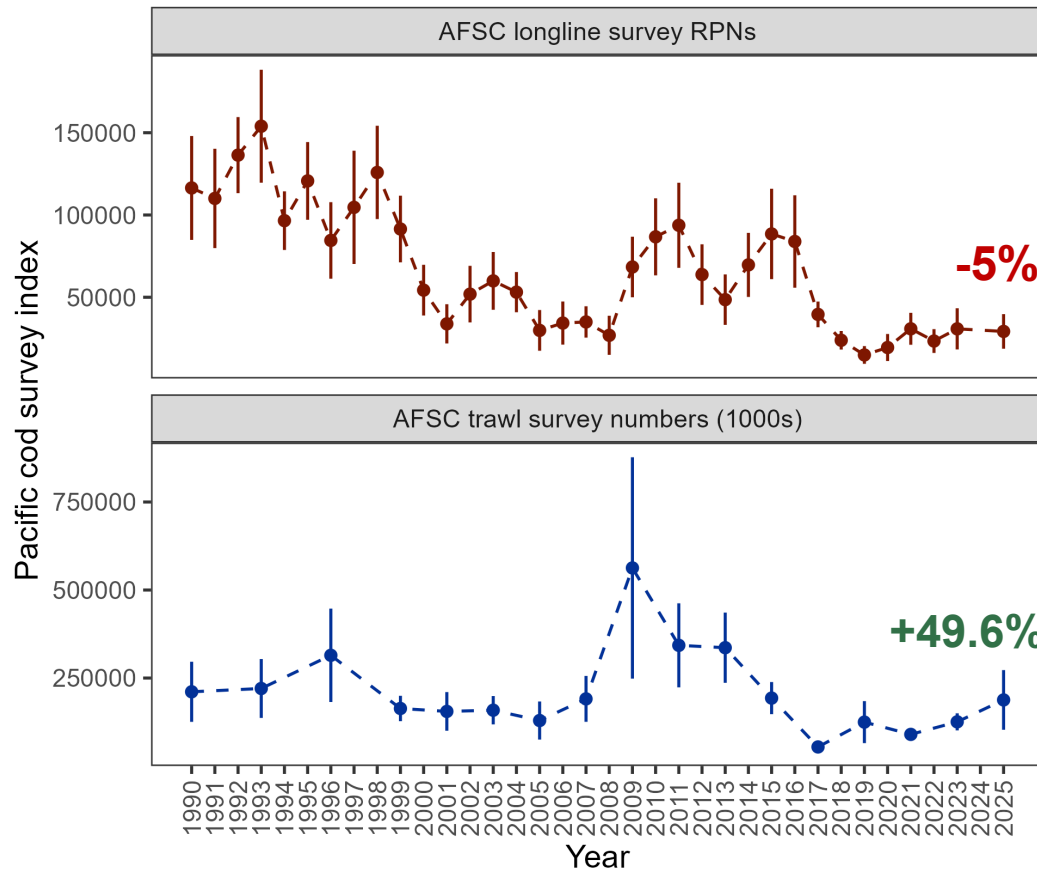


DATA - CATCH

- Nothing inconsistent in catch, highlight CGOA Jig fishery (through Dec 8, 2025)

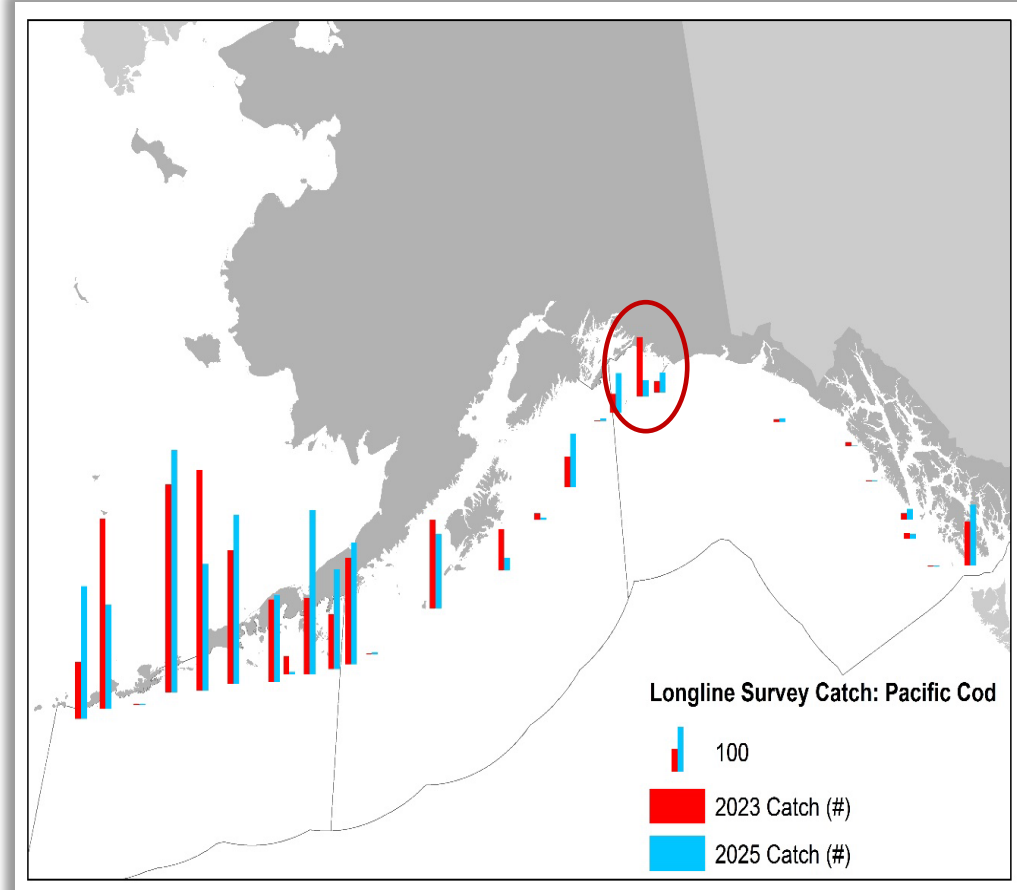


DATA – AFSC SURVEY INDICES



DATA – 2025 AFSC LONGLINE SURVEY

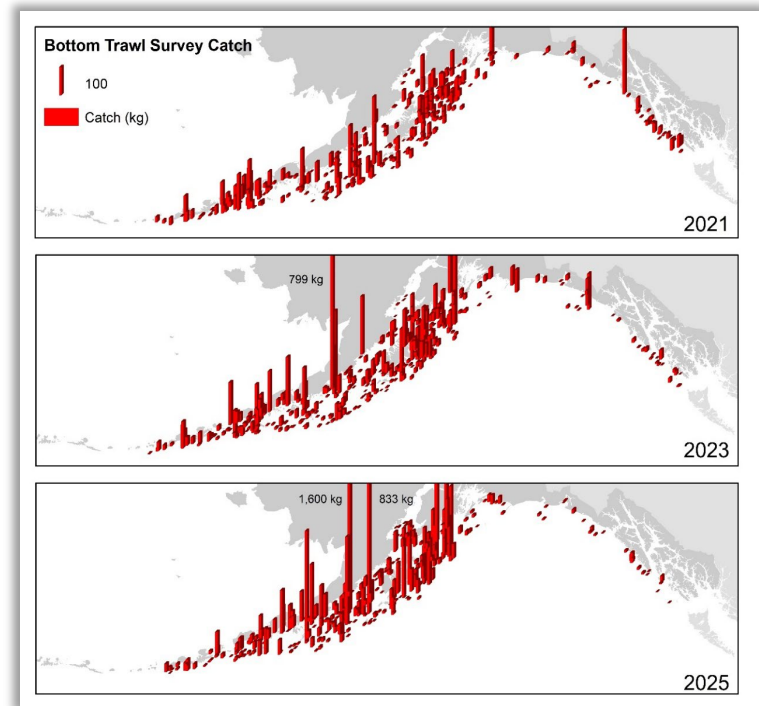
- **Bottom line:** a single station from a subregion with disproportionate area size relative to number of stations drove the decline in the RPN index
- Will be re-evaluating this index in future assessments



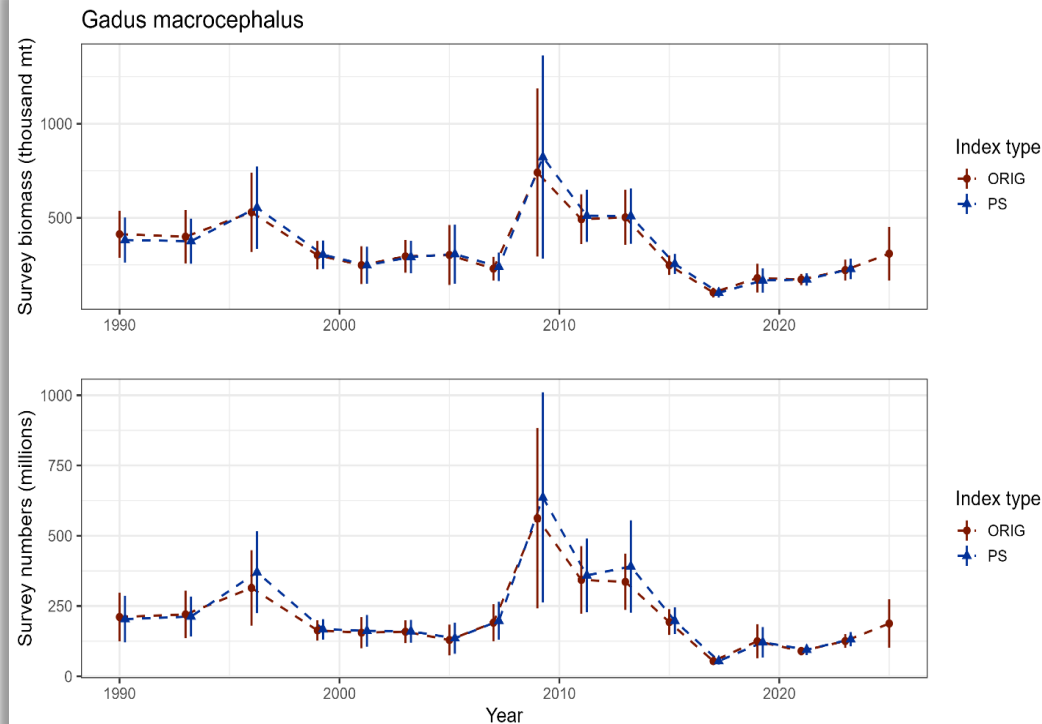
DATA – 2025 SURVEY CONTEXT

Was there an influence of the GOA bottom trawl survey restratification on 2025 results?

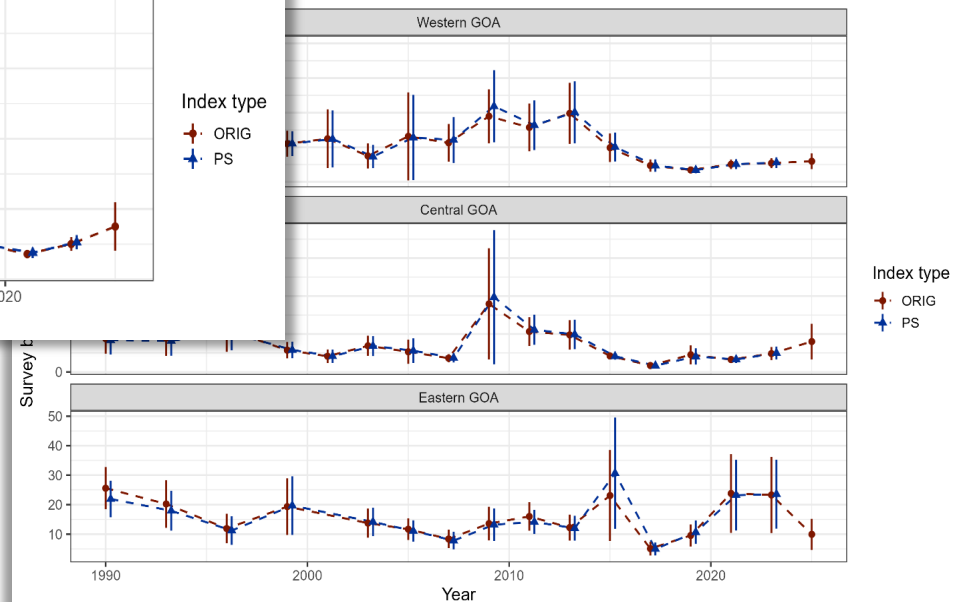
- Design implemented in 2025 has been extensively investigated and reviewed
- Using historical data, what would have happened if it were collected following new survey design? (Appendix 2.2)
 - *Does the restratification of the survey design induce any bias?*



DATA – AFSC BOTTOM TRAWL SURVEY



■ Results indicate minor differences between time-series



DATA – AFSC BOTTOM TRAWL SURVEY

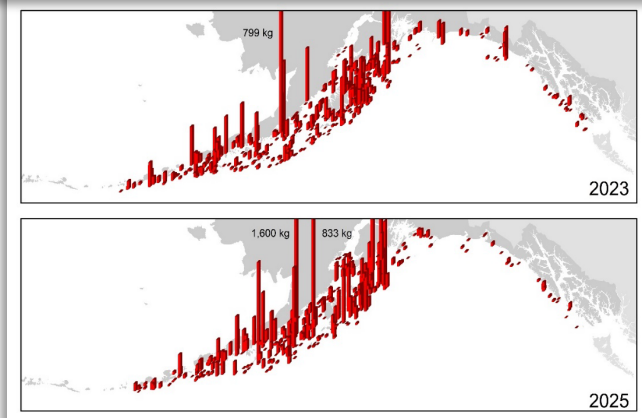
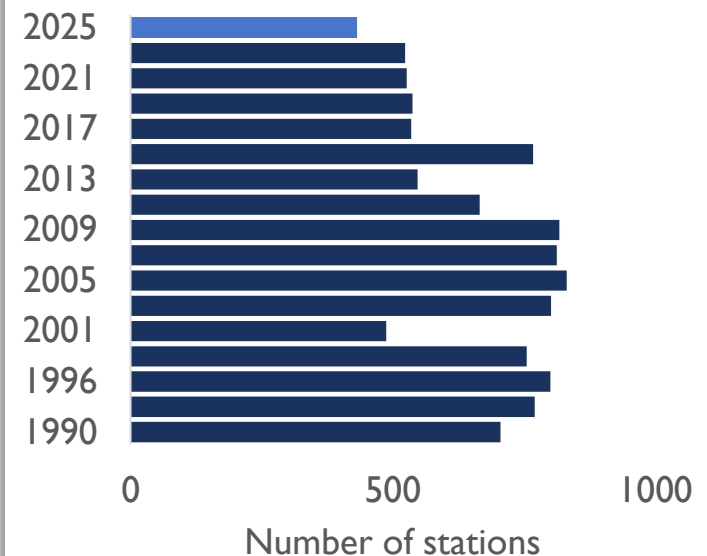
*Was there an influence of the GOA bottom trawl survey restratification on 2025 results? **NO***

- The results from the 2025 survey continue trends we have recently observed and estimates that are within historical ranges
- This analysis was not intended to create a new time-series of historical data, rather, to compare and verify that the 2025 survey restratification did not have an unintended consequence on historical data
- GOA cod is one of best test cases, sampling variability for stocks that are more difficult to sample (i.e., rockfish) will remain a factor, regardless of restratification of survey



DATA – AFSC BOTTOM TRAWL SURVEY

- Fewest stations in 2025
AFSC bottom trawl survey since 1990
- Can have an effect on index uncertainty, but uncertainty largely driven by catch variability across stations
- Will have much more of an effect on variability in index estimate for ‘zero-inflated’ species than restratification



RISK TABLE

Risk table guiding principle: what are the risks, external to the stock assessment, to the recommended 2026 ABC?

<i>Assessment-related Considerations</i>	<i>Population Dynamics Considerations</i>	<i>Ecosystem Considerations</i>	<i>Fishery-informed Stock Considerations</i>
Level 1: Normal	Level 1: Normal	Level 2: Increased Concern	Level 1: Normal
Model 24.0 does not have a concerning retrospective pattern and fits the available data well	Stock continues to experience historically low spawning biomass coupled with below average recruitment	Prolonged warm ocean temperatures throughout the water column in 2025, and concerns of prey base availability, may adversely impact adult Pacific cod biological status in 2026.	Fishery performance indicators are consistent with previous years



RISK TABLE

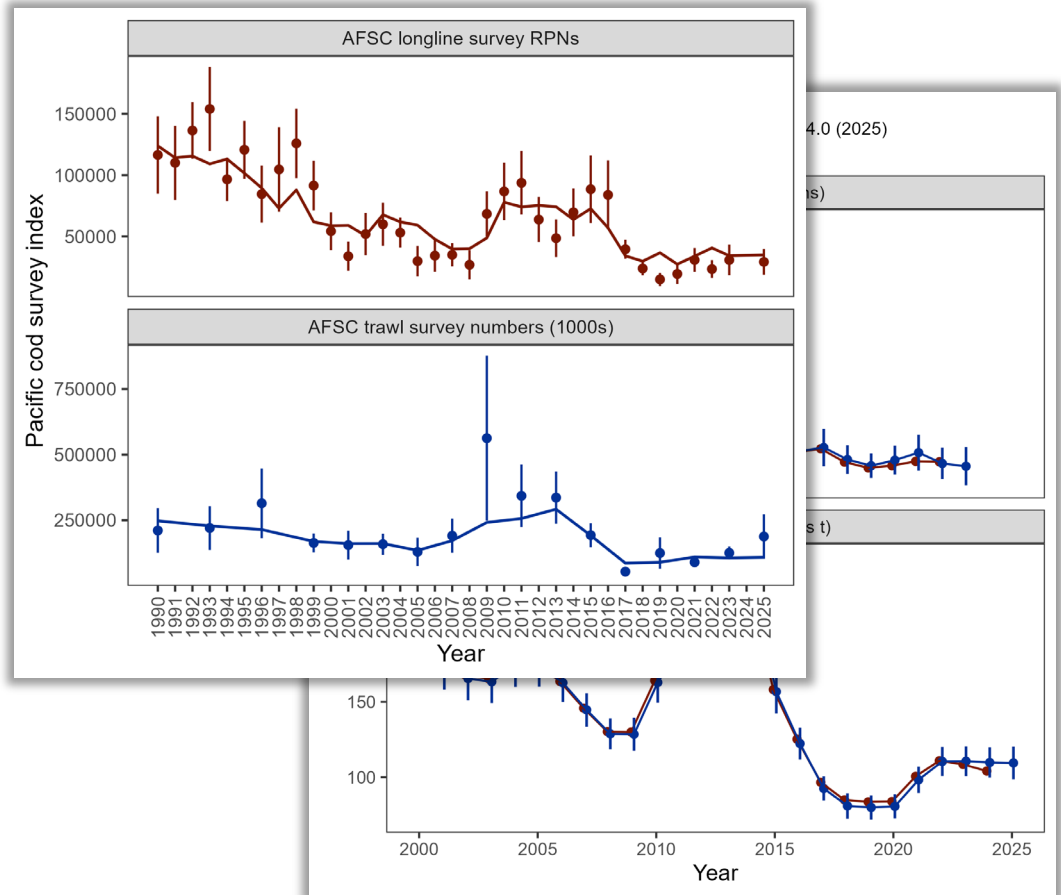
Should the 2026 ABC be reduced from maximum?

- Thinking through this question:
 - Data synthesis
 - **Model evaluation**
 - Projection considerations
 - Stock status and HCR
 - **Have we been here before, and, what happened?**

RISK TABLE

Model evaluation

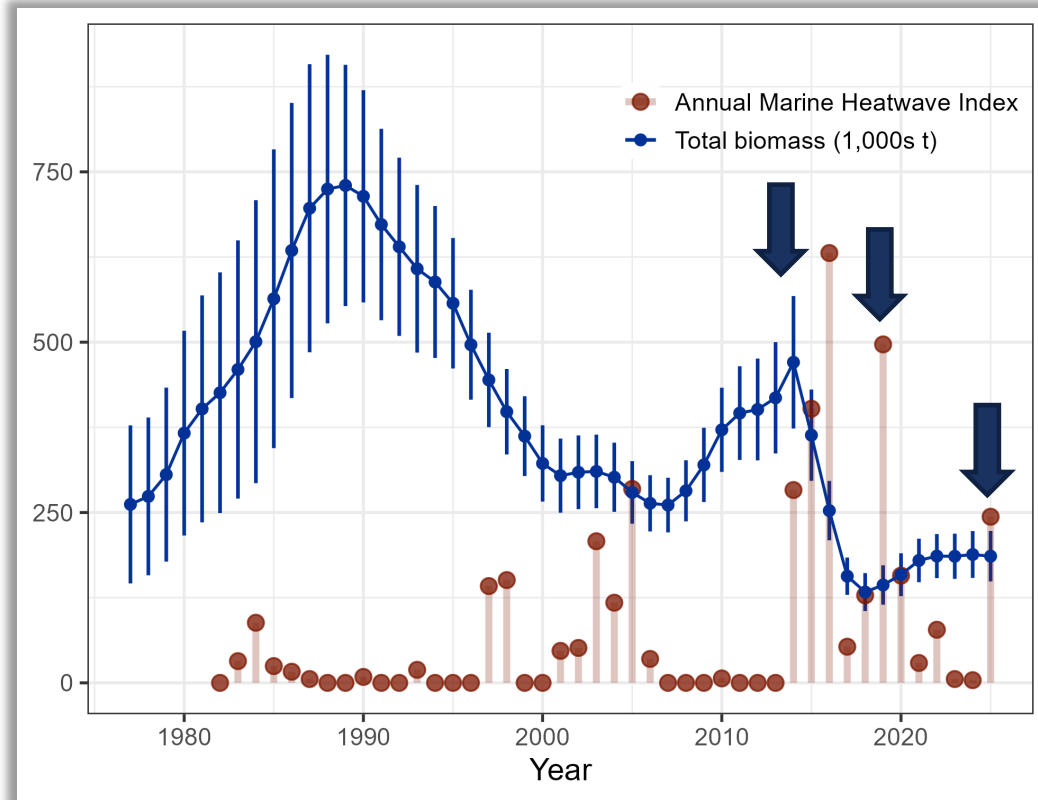
- Model does not reflect increase in available index data
- The increase in 2025 from Model 24.0 is the result of stable estimates compared to decreasing trend estimated in 2024



RISK TABLE

Have we been here before, and, what happened?

- 2014: severe marine heatwave follows, pop'n collapses
- 2019: marine heatwave decreases after, notable differences in assessment model since 2014, fishery closed in 2020



RISK TABLE

Should the 2026 ABC be reduced from maximum?

- Summary: There are several aspects of the current stock assessment that mitigate risk, including catch assumptions in projections, stock status and the HCR, and model development over time
- But, do these serve to mitigate the risk identified for the 2026 ABC?
 - It is not clear to what extent increased risk highlighted from environmental conditions in 2025 is mitigated by stock assessment
 - It remains unknown if environmental conditions in 2025 will persist into 2026

RISK TABLE

Should the 2026 ABC be reduced from maximum?

- An additional buffer to the 2026 ABC could be considered, however,
 - There is no quantitative method available with clear objectives from which to derive the additional buffer, it remains a subjective decision
 - There is no understanding of any measurable risk reductions that have occurred due to additional buffers that have historically been implemented across stocks

ABC RECOMMENDATION

Quantity	As estimated or <i>specified last</i> year for:		As estimated or <i>specified this</i> year for:	
	2025	2026	2026	2027
<i>M</i> (natural mortality rate)	0.49*	0.49*	0.5*	0.5*

Should the 2026 ABC be reduced from maximum?

Taken together, a reduction from maximum ABC in 2026 is not recommended, while at the same time acknowledge the ***increased risk*** associated with this recommendation

F_{ABC}	0.46	0.43	0.54	0.47
OFL (t)	38,688	36,459	49,782	38,812
maxABC (t)	32,141	30,193	41,520	32,209
ABC (t)	32,141	30,193	41,520	32,209
Status	As determined <i>last</i> year for:		As determined <i>this</i> year for:	
	2023	2024	2024	2025
Overfishing	No	n/a	No	n/a
Overfished	n/a	No	n/a	No
Approaching overfished	n/a	No	n/a	No

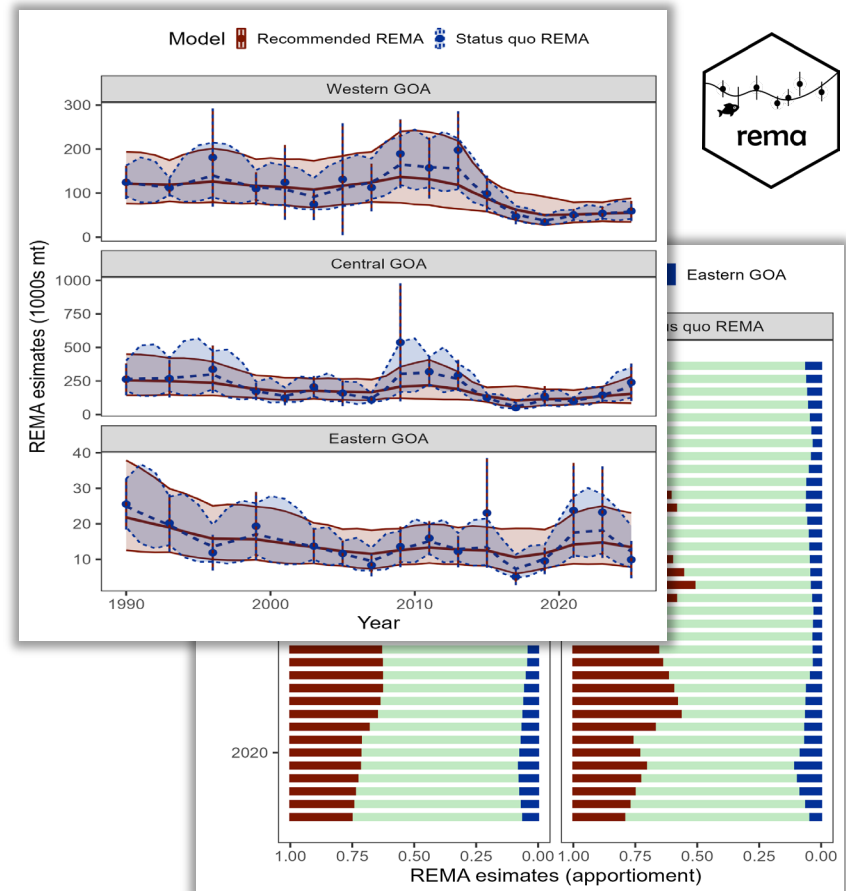
*Base natural mortality M varies between 0.5 and 0.84

** Assumed 2025 catch to be the 2025 ABC. For 2027 projections the 2026 catch was assumed to be at the projected ABC.



APPORTIONMENT

- Recommended apportionment changes:
 1. Stabilize estimates of subregion biomass and uncertainty to be reflective of assessment model estimates of biomass,
 2. Provide apportionment estimates that do not result in drastic and improbable shifts in distribution that are inconsistent with our understanding of cod life history and movement



APPORTIONMENT

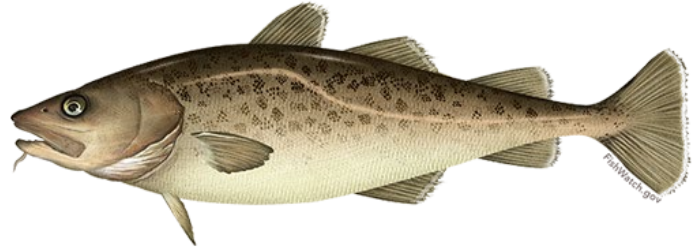
- Comparison among adopted apportionment from 2025, status quo apportionment, and recommended apportionment:

	Western	Central	Eastern	Total
Previous apportionment	27.1%	63.8%	9.1%	100%
2025 BRD	8,710	20,506	2,925	32,141
Status quo apportionment	20.6%	75.1%	4.3%	100%
2026 BRD	8,553	31,182	1,785	41,520
Recommended apportionment	24.8%	69.2%	6%	100%
2026 BRD	10,297	28,732	2,491	41,520

QUESTIONS?



EXTRA SLIDES



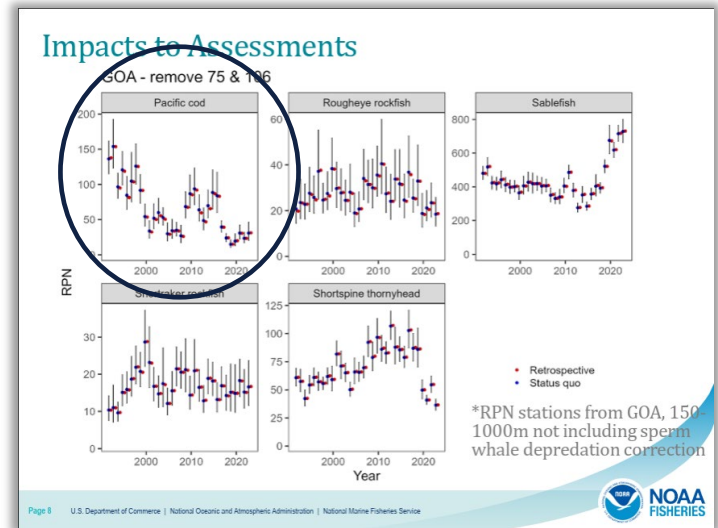
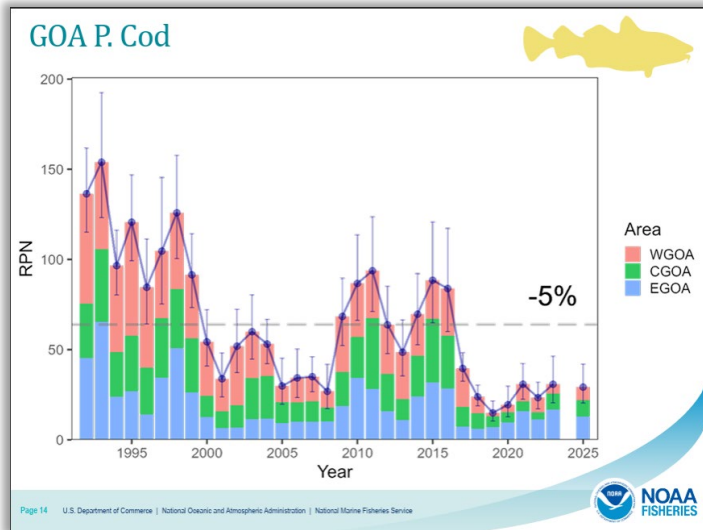
DATA – 2025 SURVEY CONTEXT

- Adding context to the survey results:
 1. Why was there an observed decline in longline survey RPNs that is inconsistent with trawl survey abundance?
 2. Was there an influence of the GOA bottom trawl survey restratification on 2025 results?

DATA – 2025 AFSC LONGLINE SURVEY

1. Why was there an observed decline in longline survey RPNs?

- Not due to dropping stations in survey redesign



- Subregion change since 2023:
WGOA **+40%**, CGOA **+4.8%**,
EGOA **-23.6%**

- This decline in GOA-wide RPN due to EGOA



DATA – 2025 AFSC LONGLINE SURVEY

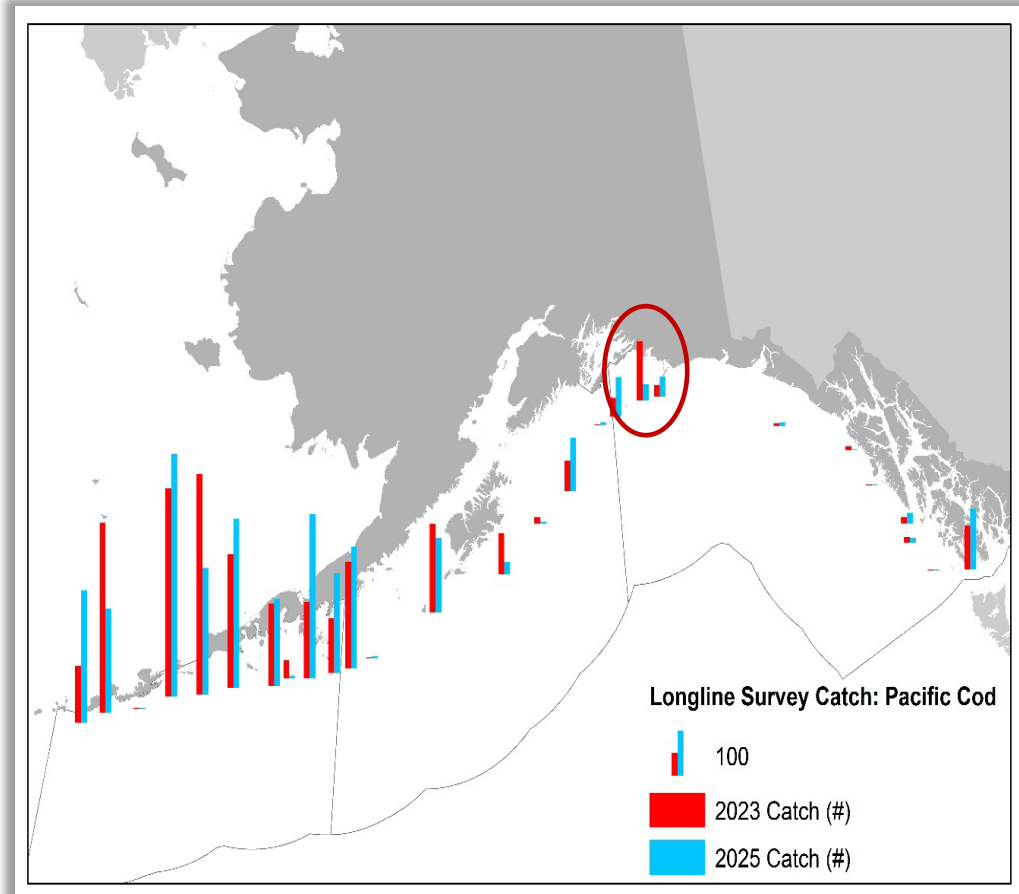
1. *Why was there an observed decline in longline survey RPNs?*
 - RPN calculation reminder: mean catch per hook (across skates within a strata) multiplied by stratum area size (stratum are 150m and deeper)
 - Drilling down into EGOA:
 - Why is the EGOA RPN so large in comparison to WGOA and CGOA?
 - What was the reason for the EGOA RPN decline?

DATA – 2025 AFSC LONGLINE SURVEY

- Drilling down into EGOA RPN: *Why so large?*
 - Since 2020 77% of cod in RPN index caught in 151-200m stratum (remainder caught in 201-300m stratum; *note: 1.3x more cod caught in 101-150m compared to 151-200m*)
 - Area size for 151-200m in EGOA 4x larger than WGOA and 3x larger than CGOA
 - Large area size has downstream effect on RPN resulting in >33% of GOA RPN within EGOA across time-series (*AFSC bottom trawl survey observes <5% of cod abundance in EGOA*)

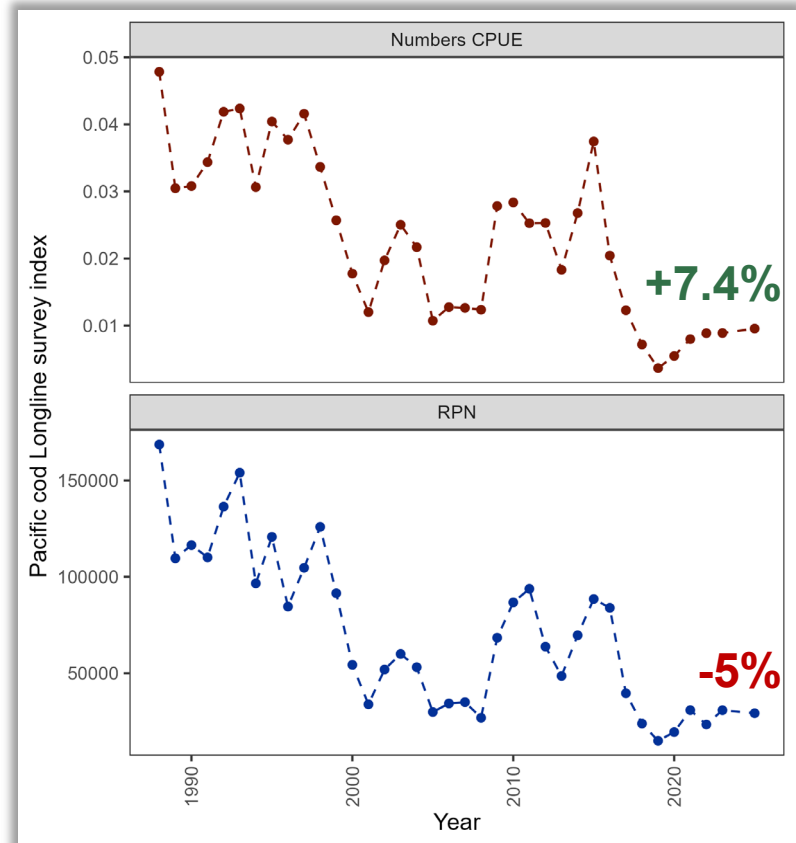
DATA – 2025 AFSC LONGLINE SURVEY

- Drilling down into EGOA:
 - Compared to 2023, only one station's catch of cod declined
 - #90: 2023 catch largest since 1993, 6.5x larger than avg since 2000



DATA – 2025 AFSC LONGLINE SURVEY

- Removing area size effect, numbers CPUE increased by more than 7%
- **Bottom line:** a single station from a subregion with disproportionate area size drove the decline in the RPN index



DATA – AFSC BOTTOM TRAWL SURVEY

Using real data, what would happen if historical data collected following new survey design?

- Not a straight-forward question to answer
- Primary difficulty to overcome is that historical stations were chosen within a grid cell with selection probabilities based on old strata
- This means that you must reweight stations after post-stratifying into new strata so that old strata with higher number of stations don't 'overwhelm' old strata with lower number of stations upon combining within the new strata

DATA – AFSC BOTTOM TRAWL SURVEY

- Upon post-stratification within new strata, before you calculate design-based indices, you must re-weight to account for the new selection probabilities, otherwise you will introduce bias
 - Can think of the weight as being the inverse of the selection probability – the higher the selection probability in the original design the lower the weight, and visa versa
- The point: must account for this in post-stratifying samples so that we can determine whether the survey redesign has unintended consequences (i.e., bias) using real-world data

DATA – AFSC BOTTOM TRAWL SURVEY

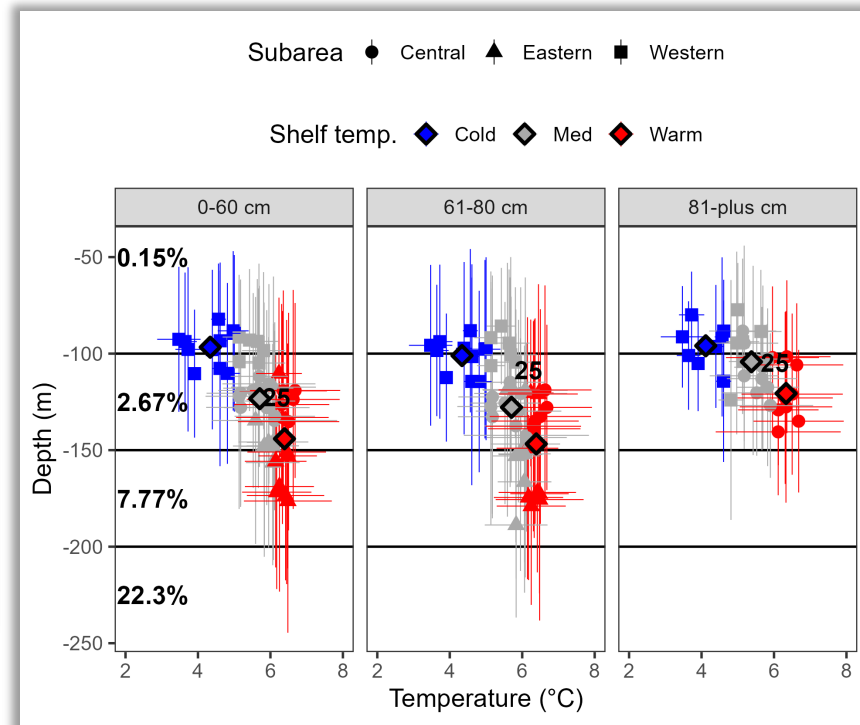
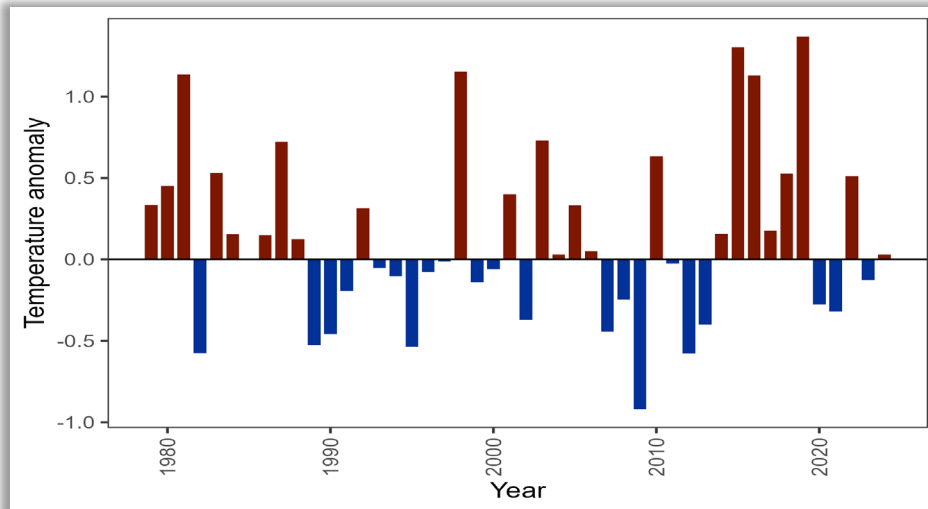
Using real data, what would happen if historical data collected following new survey design?

- Utilized survey R-package “Analysis of complex survey samples”
 - `survey::postStratify()` function follows Rao et al (2002 – analyzing survey data using post-stratification) with variance estimates following Valliant (1993), built upon Horvitz-Thompson estimator
- Steps taken in analysis:
 1. Reclassified historical stations within new 2025 stratum boundaries
 2. Computed selection probabilities and re-weighted station observations
 3. Compute design-based index



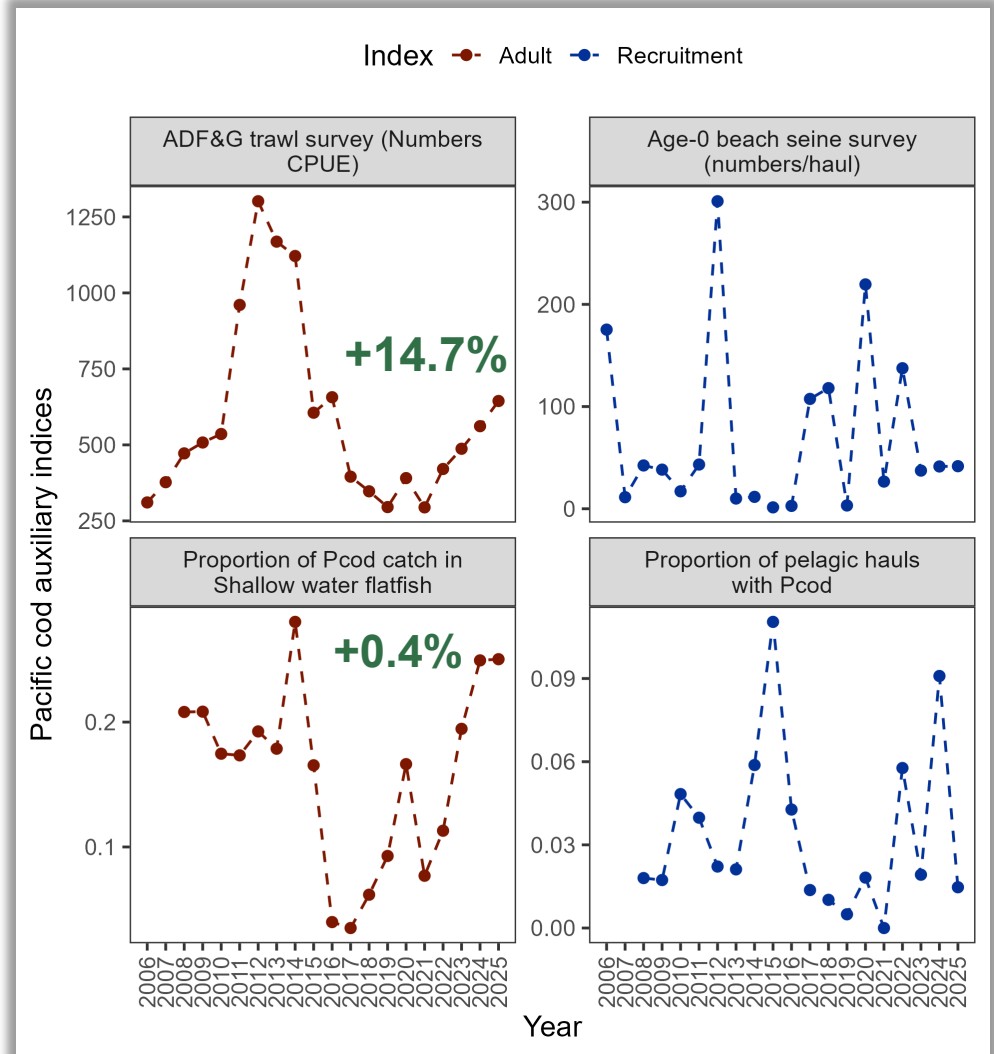
DATA – ENVIRONMENTAL

- 2025 CFSR not available – covariate not updated (new index will be used in next full assessment)
- Warmer bottom temps in bottom trawl survey associated with deeper depth on average



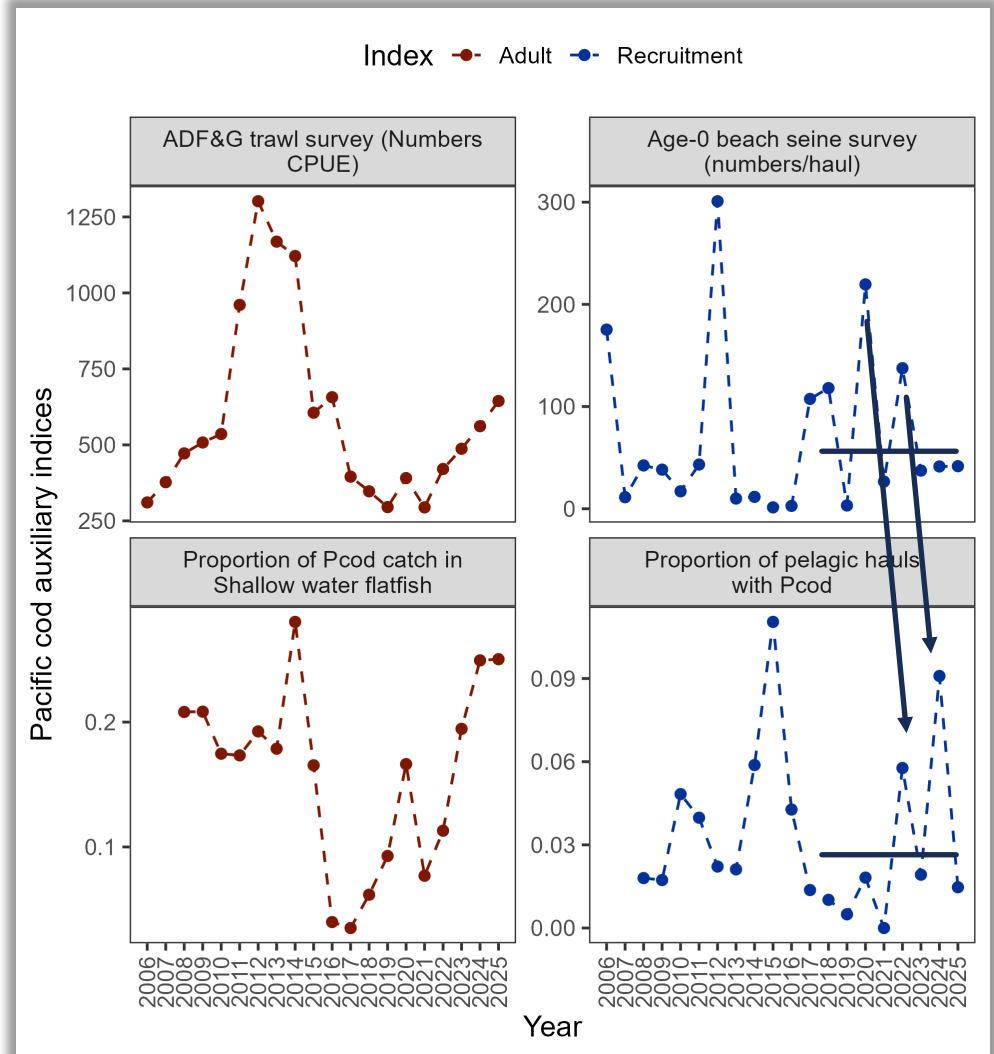
DATA – AUXILIARY INFORMATION

- Adult indices:
 - ADF&G Numbers CPUE increased
 - Proportion of cod in SWF catch stayed at >0.2



DATA – AUXILIARY INFORMATION

- Recruitment indices:
 - Beach seine 2023-2025 below average
 - 2025 below avg in pelagic hauls with cod (age-2)
- Both agree with above average 2020 and 2022 year classes



RISK TABLE

- Changes from 2024:
- Pop'n dynamics considerations: historically low spawning biomass, below average recruitment, and below average spawning conditions in 2025 are cause for concern.
- However, our understanding of spawning biomass and recruitment comes from assessment model, which, by definition, includes these considerations in any ABC recommendations
- Below average spawning conditions in 2025 will likely lead to poor 2025 year-class, but, this has little effect on 2026 ABC
- While there remains concern over the pop'n dynamics of this stock, reduce the risk level to 1 because these risks are not external to the stock assessment

RISK TABLE

- Changes from 2024:
- Ecosystem considerations: warmer than average ocean conditions in 2025, decrease in adult body condition
- Change from risk level 1 to 2 increased concern

Should the 2026 ABC be reduced from maximum?

- Given our history with this stock, we must think critically about any recommendations that could have unintended impacts on the stock

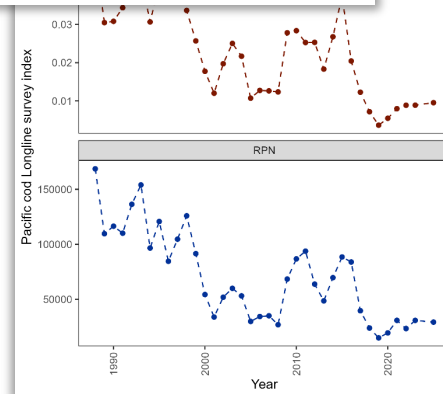
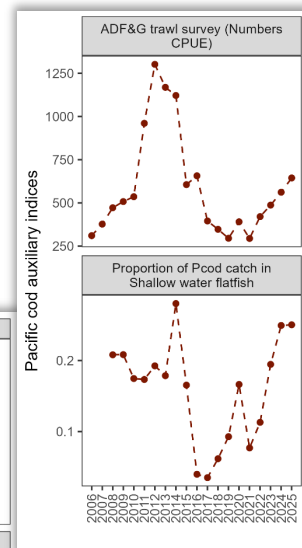
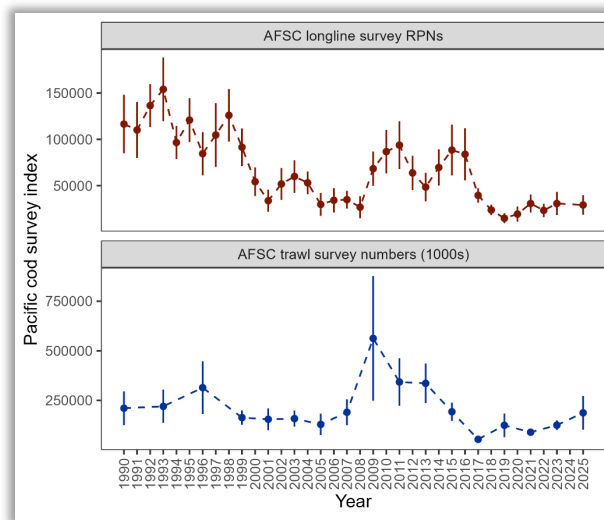
RISK TABLE

Should the 2026 ABC be reduced from maximum?

- Data synthesis:

- Available population index data indicates an increase in the population since 2023/2024

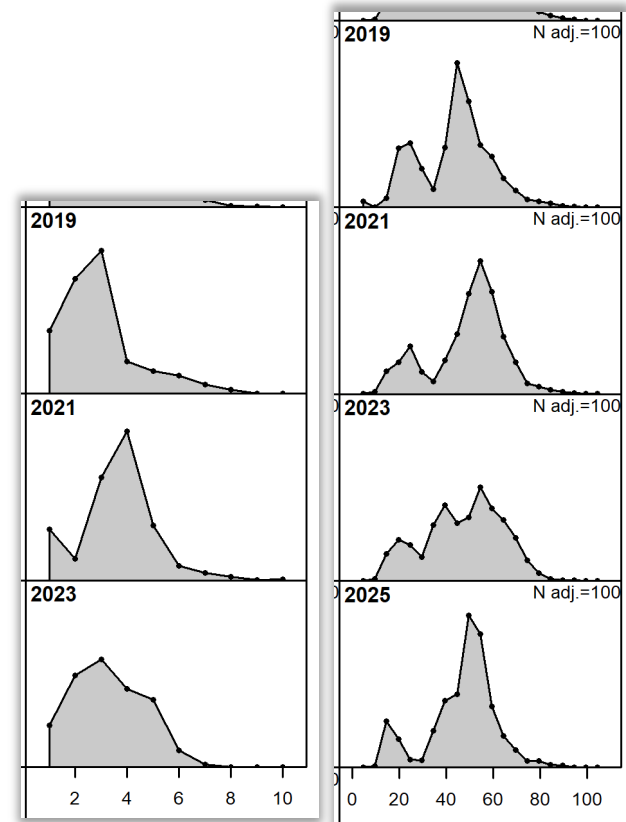
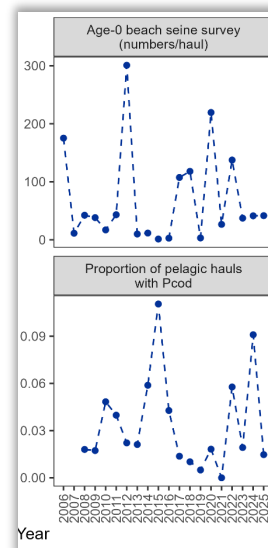
- AFSC longline survey exception, but, we've discussed why this result occurred, raw catch rates indicates slight increase



RISK TABLE

Should the 2026 ABC be reduced from maximum?

- Data synthesis:
 - Available AFSC trawl survey age and length composition consistently indicates age-1 – age-3 fish in population
 - Beach seine and pelagic catch indicate young fish in system, particularly 2020 & 2022 year classes

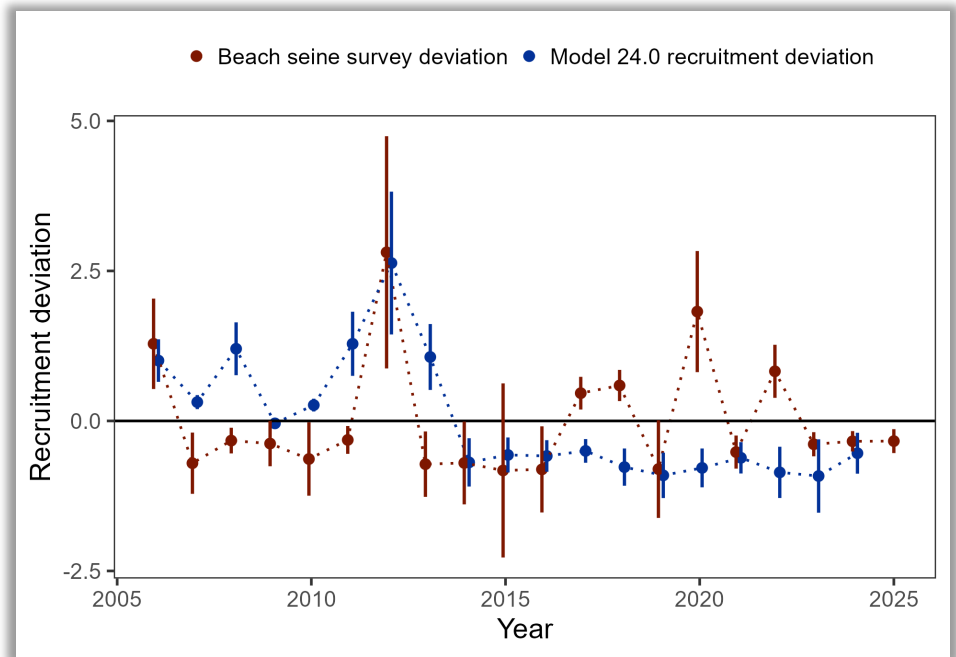


RISK TABLE

Should the 2026 ABC be reduced from maximum?

- Model evaluation:

- Model continues to estimate below average recruitment, but, these estimates do not compare to available data well

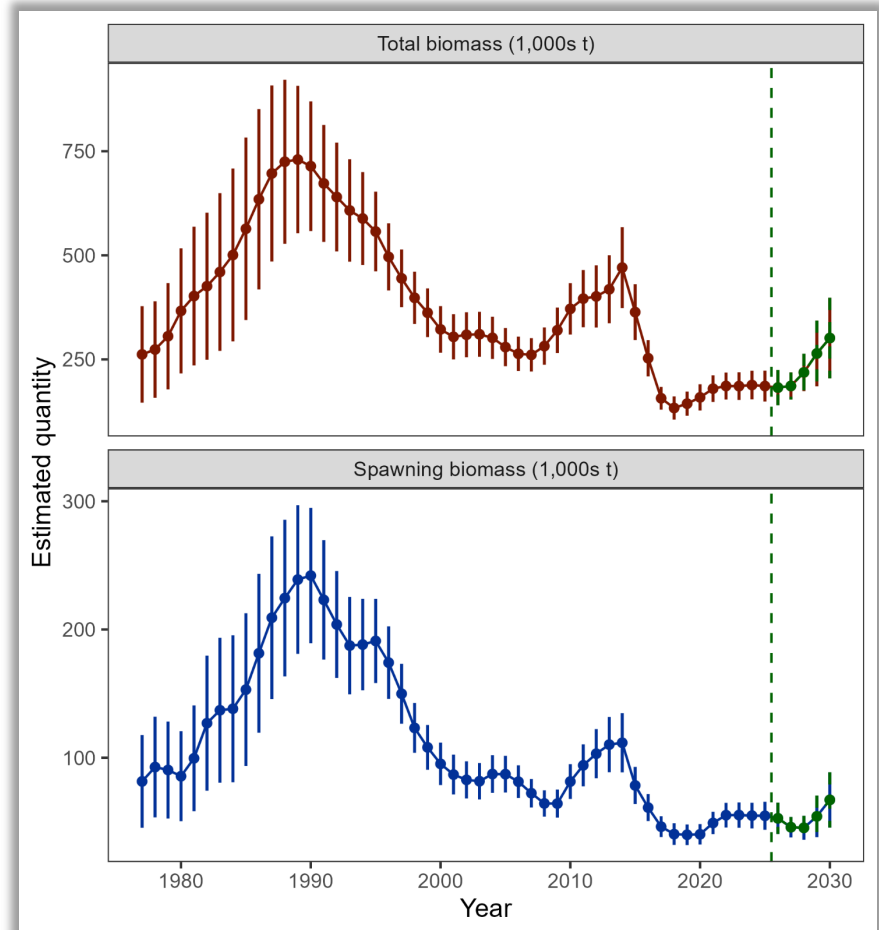


- There is a possibility that the model estimates are misrepresenting recruitment

RISK TABLE

Should the 2026 ABC be reduced from maximum?

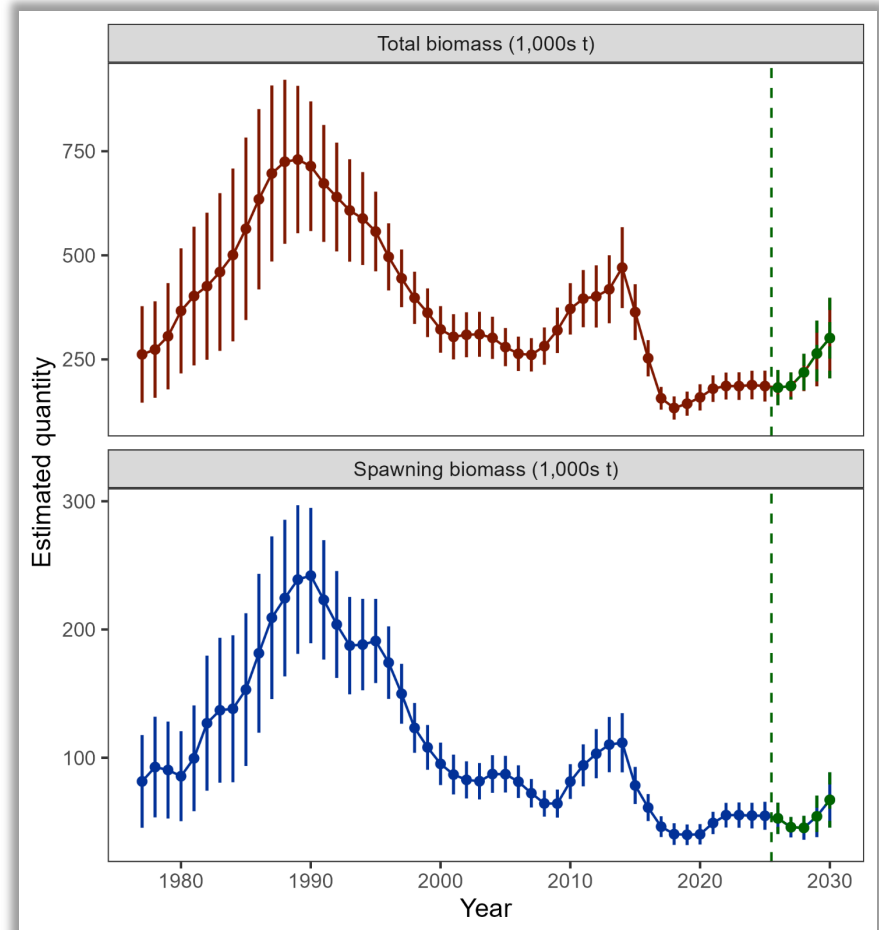
- Projection considerations:
 - Continue to note the average recruitment concern in projected biomass, when recent recruitment estimated to be below average since 2014; this potentially results in overly-optimistic long-term projections, but does not have large effect on 2026 ABC



RISK TABLE

Should the 2026 ABC be reduced from maximum?

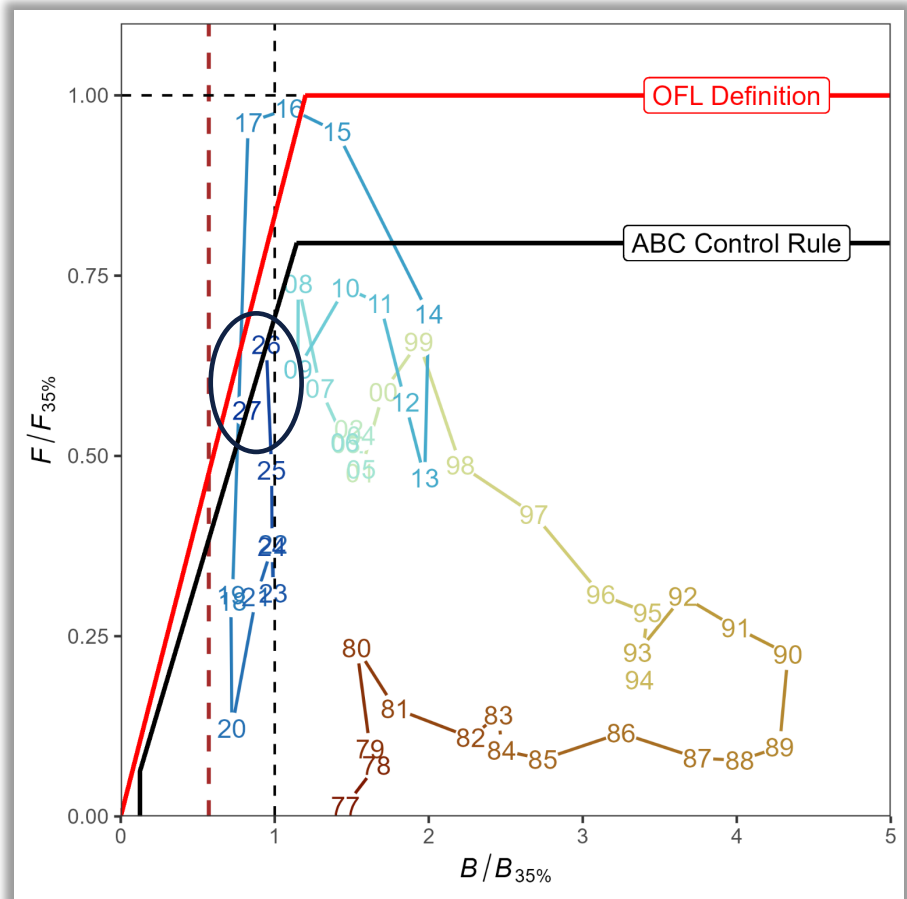
- Projection considerations:
 - Assumption of utilizing full ABC in terminal year of model for projections reduces 2026 recommended ABC by 5% as compared to using observed catch proportions since fishery closure in 2020



RISK TABLE

Should the 2026 ABC be reduced from maximum?

- Stock status and HCR:
 - Stock is currently in Tier 3b, on the ramp of the HCR
 - Based on the HCR and projected stock status, $F_{40\%}$ reduced by 18% to obtain 2026 F_{ABC}



APPORTIONMENT

- Two REMA model configuration changes recommended for apportionment (BRD)
 1. Estimate single process error parameter across GOA subregions
 2. Estimate additional observation error parameter for GOA bottom trawl survey
- These REMA model configurations previously adopted for GOA Thornyhead stock complex (Echave et al. 2022, Siwicke et al. 2024)

